

## Combinatorics PhD Exam, May 1991

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Do 9 out of 12 problems. Show all of your work.

1. How many  $k$ -tuples  $(A_1, A_2, \dots, A_k)$  of nested subsets of an  $n$ -set are there, where  $A_1 \subseteq A_2 \subseteq \dots \subseteq A_k$ ?
2. How many words of length  $n$  are there in the alphabet  $(a, b, c)$  with no two adjacent  $a$ 's? With no two adjacent letters equal?
3. Find the generating function for the sequence  $\{a_n\}$  given by the recurrence

$$a_n = a_{n-1} + 6a_{n-2} + 2^n$$

where  $a_0 = 1$  and  $a_1 = 3$ .

4. Bipartite graphs:

(a) Determine for which values of  $m$  and  $n$  the complete bipartite graph  $K_{mn}$  is

- (1) planar;
- (2) Eulerian;
- (3) Hamiltonian.

(b) Prove: A plane graph where every face has an even number of edges must be bipartite.

5. Prove that

$$\sum_{k=1}^n S(n, k) x(x-1) \cdots (x-k+1) = x^n.$$

6. Use Möbius inversion to find the number of integers between 1 and  $n$  and coprime to  $n$ , given the prime factorization of  $n$ .
7. Prove or disprove: There is a matching from  $V$  to  $W$  in a bipartite graph if, for some fixed  $k$ , there are  $k$  or more edges incident with each vertex of  $V$  and  $k$  or fewer edges incident with each vertex of  $W$ .
8. Let  $A_{m,n}$  be the collection of all sets  $X = \{N_1, N_2, \dots, N_k\}$ , where  $k$  is a non-negative integer, and for each  $i \leq k$ ,  $N_i$  is a positive integer divisor of  $N = 2^m 3^n$ , and where  $N_i \mid N_j$  only if  $i = j$ . Define a partial order on  $A_{m,n}$ , where  $X = \{N_1, N_2, \dots, N_k\}$ ,  $X' = \{N'_1, N'_2, \dots, N'_l\}$ , by  $X \preceq X'$  if and only if  $N_i \in X \Rightarrow$  there exists  $N'_j \in X'$  with  $N_i \mid N'_j$ . Prove that  $A_{m,n}$  is a distributive lattice, and determine its poset of join-irreducible elements.
9. Let  $I$  be a collection of subsets of  $E$ , where  $E$  is finite, and suppose that  $I \neq \emptyset$  and that  $Y \in I$ ,  $X \subseteq Y \Rightarrow X \in I$ . Let  $c : E \rightarrow \mathbb{R}^+$  assign a positive real number to each element of  $E$ . Consider the optimization problem:  
Find  $\max_{X \in I} \sum_{x \in X} c(x)$   
Prove that the greedy algorithm solves this problem if and only if  $I$  is the collection of independent sets of some matroid on  $E$ .
10. State the MacWilliams identity, and use it to compute the weight polynomial of the  $(15, 11)$  Hamming binary code.

11. A  $(v, k, \lambda)$ -design  $\Sigma$  is an incidence structure consisting of a set  $S$  of  $v$  points and a collection of  $k$ -subsets of  $S$  called blocks, such that every pair of points of  $S$  is contained in precisely  $\lambda$  blocks.

Prove that, in the range  $30 \leq v \leq 45$ , a  $(v, 6, 1)$ -design exists if and only if  $v = 31$ .

12. Use the Hall Multiplier Theorem

- (i) to prove the non-existence of a cyclic  $(31, 10, 3)$  difference set.
- (ii) to find a cyclic  $(31, 6, 1)$  difference set.